

Week 03 Task 7

Malware Types & Common Attacks

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1) Executive Summary

This task focused on understanding common cybersecurity threats, malware, and attack techniques. I studied different types of malware, including viruses, worms, Trojans, ransomware, spyware, and rootkits, along with how ransomware spreads and its impact. I also explored common attacks such as phishing, DoS/DDoS, MITM, and brute-force attacks, including their detection signs. For practical learning, I completed the TryHackMe **Introduction to Cyber Security/Attacks** and **Common Attacks** rooms and studied **Vulnerabilities 101**. These activities helped me understand common attack methods, how vulnerabilities can be identified and exploited, and how suspicious activity can be detected.

2) Malware Types

Malware is a malicious software intentionally designed to gain unauthorized access of someone's system, server, network in order to harm, disrupt operations, steal sensitive data or exploiting system resources to gain access as a normal or admin user. Commonly used malwares are described below.

- Virus
- Worm
- Trojan
- Ransomeware
- Spyware
- Rootkits

2.1) Virus

A malicious program that attaches itself to legitimate, executable files and spreads when those files are executed, corrupting or deleting data along the way.

2.2) Worm

A standalone malware program that replicates itself automatically and spreads across network connections without needing human interaction or a host file.

2.3) Trojan

A malicious file or program disguised as legitimate software to trick users into downloading and executing it, often creating a backdoor for attackers.

2.4) Ransomeware

A type of malware that encrypts a victim's files or locks their system, demanding a money or ransom from the user in exchange for the decryption key.

2.5) Spyware

Secretly installed software that monitors user activity, capturing sensitive data like keystrokes, login credentials, and browsing habits without the user's knowledge. Can be used for spying as an attacker. It can also be used as a parental control software.

2.6) Rootkit

A covert collection of tools designed to gain administrator-level (root) access to a system while hiding its own presence and other malicious processes from detection.

3) Common Cyber Attacks

3.1) Phishing

Phishing is a social engineering attack in which an attacker attempts to trick a user into providing sensitive information such as login credentials, financial information, or personal data. Attackers commonly impersonate trusted individuals, companies, or services and use emails, messages, or fake websites to make the request appear legitimate.

3.1.1) Detection Sign of Phishing

Some common signs of phishing include suspicious emails, urgent requests for sensitive information, misleading sender addresses, suspicious links, grammatical errors, and unusual domain names. Users should verify the sender and links before providing credentials or other sensitive information.

3.2) DoS/DDoS

A Denial-of-Service (DoS) attack attempts to make a system, server, or network service unavailable by overwhelming it with a large amount of traffic or requests. A Distributed Denial-of-Service (DDoS) attack works in a similar way but uses multiple compromised systems or devices to generate traffic toward the target, making it more difficult to handle and block.

3.2.1) Detection Sign of DDos

Common signs of a DoS/DDoS attack include unusual network traffic, slow server response, high resource usage, frequent connection failures, and service unavailability. Monitoring traffic against the normal baseline can help identify such attacks.

3.3) MITM

A Man-in-the-Middle (MITM) attack occurs when an attacker positions themselves between two users and intercepts or potentially modifies the communication without the parties being aware. This can allow the attacker to capture sensitive information such as login credentials, session data, or other transmitted information.

3.3.1) Detection Sign of MITM

Possible signs of a MITM attack include unexpected certificate warnings, unusual network behavior, redirected connections, or suspicious changes in DNS and network settings. Using HTTPS and verifying security certificates can help detect and reduce the risk of MITM attacks.

3.4) Bruteforce

A brute-force attack is an attempt to gain unauthorized access by repeatedly trying different passwords or credentials until the correct combination is found. Attackers may use automated tools to

perform a large number of login attempts.

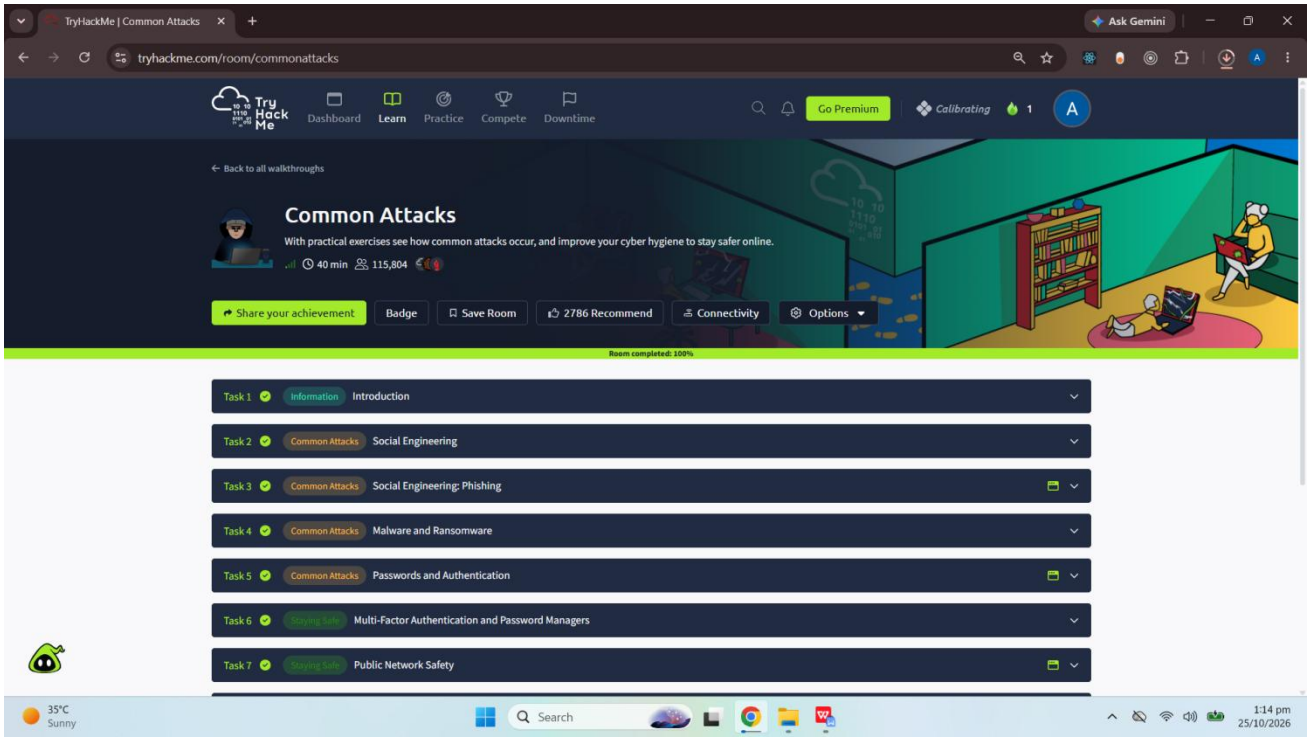
3.4.1) Detection Sign of Bruteforce

Common signs include repeated failed login attempts, multiple login attempts from the same or unusual IP address, account lockouts, and login attempts occurring at unusual times.

4) How Ransomware Spreads and Its Impact

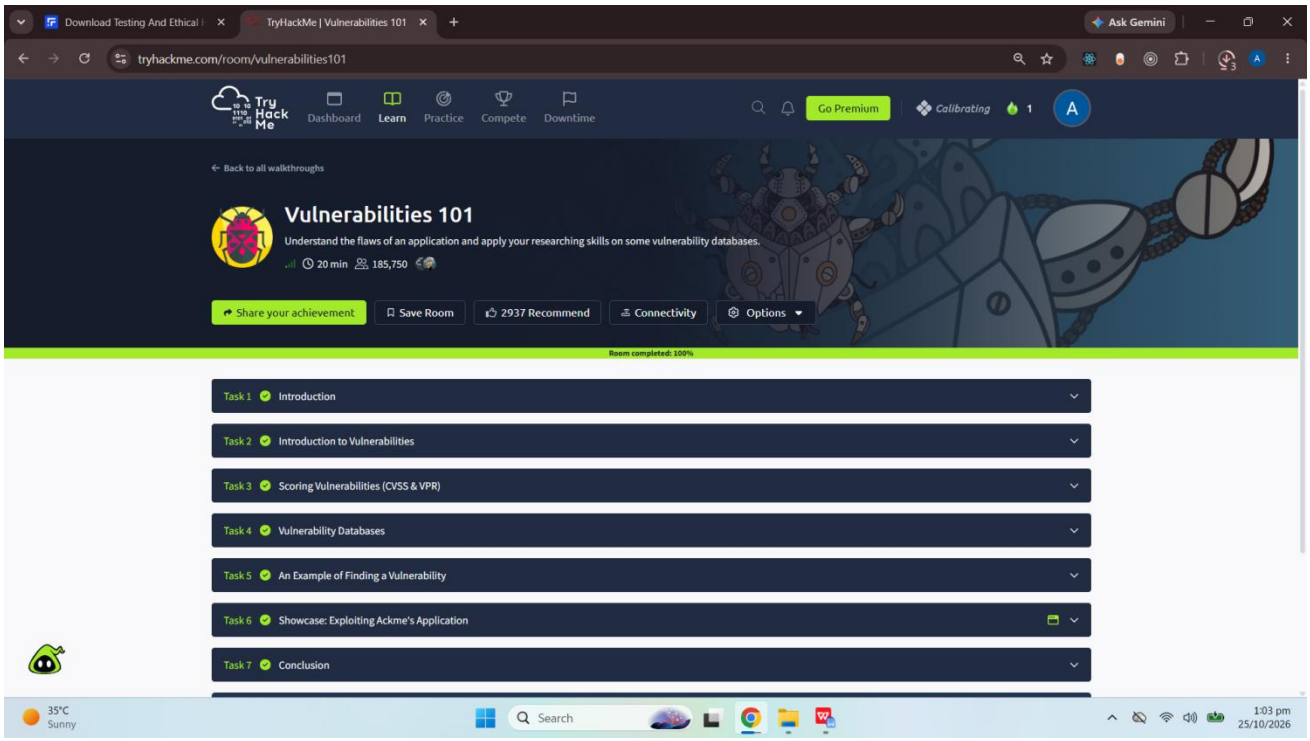
Ransomware is a type of malware that can encrypt a victim's files or lock access to a system and demand a ransom, usually in exchange for a **decryption key** or restoration of access. This can prevent users from accessing important or sensitive data and may cause significant operational and financial impact. Ransomware can spread through different methods depending on the attacker's methodology. Common methods include downloading **malicious software** disguised as legitimate applications, opening **malicious email attachments** or links, **exploiting vulnerable systems**, and using infected **USB flash drives** or other removable media. Once executed or introduced into a system, ransomware may encrypt files and, in some cases, attempt to spread to other accessible systems.

5) TryHackMe Room Common Attacks



In the above picture it is shown that I completed the **TryHackMe – Common Attacks** room, which provided practical understanding of common cybersecurity threats and attack techniques. The room covered topics including social engineering, phishing, malware and ransomware, password security, multi-factor authentication, password managers, and network safety. Through this room, I strengthened my understanding of how these attacks work and how they can be identified and prevented.

6) TryHackMe Room Vulnerabilities 101



In the above picture it is shown that I completed the TryHackMe Vulnerabilities 101 room, which provided a practical understanding of how to identify, evaluate, and search for application security flaws. The room covered topics including an introduction to vulnerabilities, scoring vulnerabilities using frameworks like CVSS and VPR, researching vulnerability databases, analyzing practical examples of finding vulnerabilities, and demonstrating a hands-on showcase of exploiting Ackme's application. Through this room, I strengthened my understanding of how software flaws are categorized, assessed for severity, and researched using industry-standard databases.

7) Conclusion

This task helped me develop a better understanding of common cybersecurity threats, malware types, and attack techniques. I learned how attacks such as phishing, brute-force, DoS/DDoS, and MITM can affect systems and how their common indicators can be identified. The practical TryHackMe rooms and Vulnerabilities 101 also improved my understanding of vulnerabilities, attack methods, and basic threat detection. Overall, this task strengthened my ability to recognize common security threats and understand their potential impact on systems and data.